

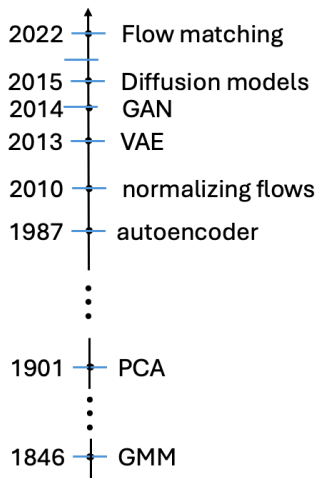
COMM114: Generative AI - Flow Matching-part 1

Guoqiang Zhang

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Flow Matching

Flow matching (FM) was invented in 2022 as a generalization of normalizing flows.



Images generated by flow matching models



Image from Flux, a particular flow matching model

Videos generated by flow matching models

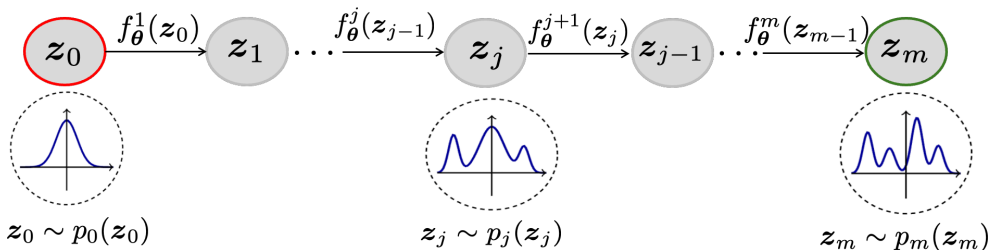


video from VideoGAM, a particular flow matching model

Recap for Normalizing Flows

Probability path over mapping index j :

- As the mapping index j increases from $j = 0$ until $j = m$, the probability density function $p_j(z_j)$ gradually transforms from a Gaussian distribution $p_0(z_0)$ to an approximation $p_m(x)$ (we let $x = z_m$) of the data distribution $p_{data}(x)$.



- As a result, we obtain a probability path $p_j(z_j)$ over the mapping index $0 \leq j \leq m$.

Objective of Flow Matching Models

- **Input:** data samples $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$.



- **Assumption:** Assume $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the data distribution $p_{data}(\cdot)$.
- **Objective of FM:** To learn a **flow** $\mathbf{X}_t^\theta \in \mathbb{R}^d$ over the time interval $t \in [0, 1]$ such that

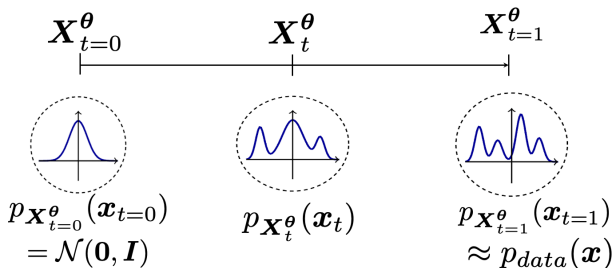
$$p_{\mathbf{X}_{t=0}^\theta}(\mathbf{x}_{t=0}) = \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad [\text{PDF of flow at } t=0] \quad (1)$$

$$p_{\mathbf{X}_{t=1}^\theta}(\mathbf{x}_{t=1}) \approx p_{data}(\mathbf{x}) \quad [\text{PDF of flow at } t=1] \quad (2)$$

- For each $t \in [0, 1]$, $\mathbf{X}_t^\theta$ is a d -dimensional continuous random variable.
- θ is the parameter of a deep neural network to be learned.

Objective of Flow Matching Models (cont.)

Probability path of a flow $\mathbf{X}_t^\theta \in \mathbb{R}^d$ over time interval $t \in [0, 1]$

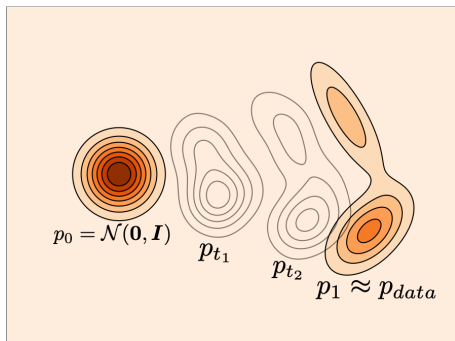


- Similarly to normalizing flow (NF), a flow matching model attempts to gradually transform a **Gaussian distribution** into an approximation of the **data distribution** as the time t increases from 0 to 1.
- Similarly to normalizing flow (NF), we obtain a probability path $p_{\mathbf{X}_t^\theta}(\mathbf{x}_t)$ over the continuous time $t \in [0, 1]$.

Objective of Flow Matching Models (cont.)

Demonstration of probability path in a 2d space

- The flow matching model attempts to gradually transform a **Gaussian distribution** into an approximation of the **data distribution** as the time t increases from 0 to 1.



Remark: In the figure above, each PDF is represented by a set of contours.

Remark: picture from [Tutorial on Flow Matching for Generative Modeling](#)

Velocity field of an FM model

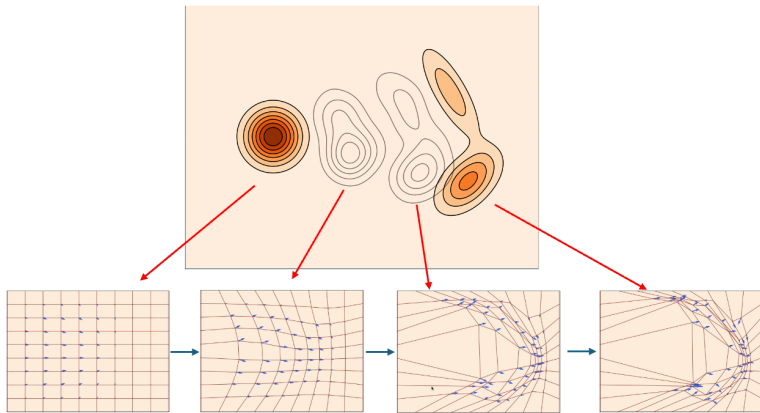
A velocity field is easier to learn:

- **Input:** data samples $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$.
- **Assumption:** Assume $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the data distribution $p_{data}(\cdot)$.
- Instead of learning the flow $\{\mathbf{X}_t^\theta \in \mathbb{R}^d | t \in [0, 1]\}$ directly, we learn a so-called **velocity field** $\mathbf{u}_t^\theta(\mathbf{X}_t^\theta)$ over $t \in [0, 1]$ from the dataset $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$.
- The velocity field $\mathbf{u}_t^\theta(\mathbf{X}_t^\theta)$ is related with the flow $\mathbf{X}_t^\theta$ via an ordinary differential equation (ODE):

$$\frac{d}{dt}\mathbf{X}_t^\theta = \mathbf{u}_t^\theta(\mathbf{X}_t^\theta) \quad \text{s. t. } \mathbf{X}_{t=0}^\theta \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad [\text{ODE}]$$

- Once the velocity field $\mathbf{u}_t^\theta(\cdot)$ is obtained, the flow $\{\mathbf{X}_t^\theta | t \in [0, 1]\}$ is the solution of the ODE.

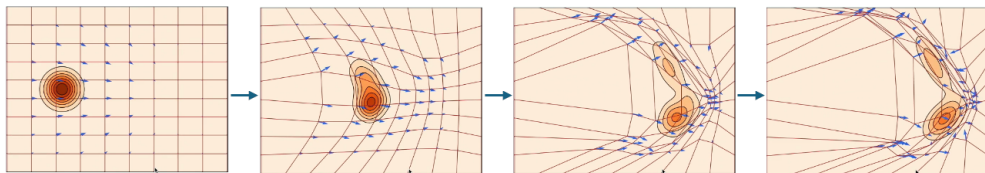
Velocity field of an FM model (cont.)



Dynamics of velocity field over $t \in [0, 1]$

- The grid points represent flow vectors.
- The **blue** arrows denote the velocities of the flow vectors.
- picture from [Tutorial on Flow Matching for Generative Modeling](#)

velocity field of an FM model (cont.)

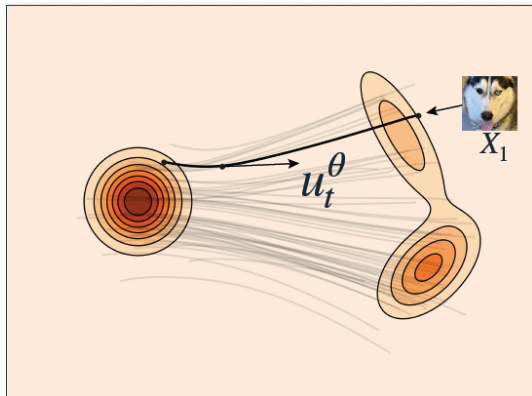


Dynamics of velocity field over $t \in [0, 1]$

- At each time t , the velocity field determines which direction and to what extent the flow evolve locally.
- picture from [Tutorial on Flow Matching for Generative Modeling](#)

Sampling from an FM model

- Once the velocity field is fixed, the probability path over $t \in [0, 1]$ is determined.
- In other words, there is a one-to-one mapping between the samples from $p_{\mathbf{X}_{t=0}^{\theta}}(\mathbf{x}_{t=0}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the samples from $p_{\mathbf{X}_{t=1}^{\theta}}(\mathbf{x}_{t=1}) \approx p_{data}(\mathbf{x})$.



- picture from [Tutorial on Flow Matching for Generative Modeling](#)

Sampling from an FM model (cont.)

Algorithm 1 Sampling from an FM model with Euler method

- 1: Input: velocity field $\mathbf{u}_t^\theta$ (a DNN model), number of steps k
 - 2: Set $t = 0$
 - 3: Set stepsize $h = \frac{1}{k}$
 - 4: Draw a sample $\mathbf{x}_0 \sim p_{init} = \mathcal{N}(0, \mathbf{I})$
 - 5: **for** $i = 1, \dots, k$ **do**
 - 6: $\mathbf{x}_{t+h} = \mathbf{x}_t + h\mathbf{u}_t^\theta(\mathbf{x}_t)$
 - 7: Set $t \leftarrow t + h$
 - 8: **end for**
 - 9: return $\mathbf{x}_{t=1}$
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Remark 1: In principle, any ODE solver can be applied to perform sampling for an FM model.

Training of velocity field

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $\mathbf{X}_1$ be $p_{data}(\mathbf{x})$ (i.e., $p_{\mathbf{X}_1}(\mathbf{x}_1) = p_{data}(\mathbf{x})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Objective function for $\mathbf{u}_t^\theta(\cdot)$:

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \mathbf{u}_t^{\text{target}}(\mathbf{X}_t) \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \frac{d}{dt} \mathbf{X}_t \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \frac{d\beta_t}{dt} \mathbf{X}_0 - \frac{d\alpha_t}{dt} \mathbf{X}_1 \right\|^2 \end{aligned}$$

Training procedure

Algorithm 2 Flow Matching Training Procedure

- 1: Input: A dataset of samples drawn from p_{data} , neural network $\mathbf{u}_t^\theta$
- 2: **while** not converge **do**
- 3: Set $\mathcal{L}(\theta) = 0$
- 4: **for** each mini-batch of data **do**
- 5: Select a sample $\mathbf{x}$ from the dataset
- 6: Sample a random time $t \sim \mathcal{U}[0, 1]$
- 7: Sample noise $\epsilon \sim \mathcal{N}(0, \mathbf{I}_d)$
- 8: Set $\mathbf{x}_t = \beta_t \epsilon + \alpha_t \mathbf{x}$
- 9: Compute loss

$$\mathcal{L}(\theta) = \mathcal{L}(\theta) + \left\| \mathbf{u}_t^\theta(\mathbf{x}_t) - \left(\frac{d\beta_t}{dt} \epsilon + \frac{d\alpha_t}{dt} \mathbf{x} \right) \right\|^2$$

- 10: **end for**
- 11: Update the model parameter via gradient descent
$$\theta \leftarrow \theta - \eta \frac{\mathcal{L}(\theta)}{d\theta}$$

- 12: **end while**
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Training of velocity field (cont.)

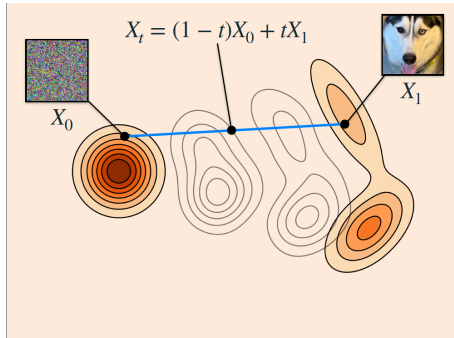
Linear functional form of α_t and β_t

- α_t and β_t are set as

$$\alpha_t = t$$

$$\beta_t = 1 - t$$

- As a result $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1 = (1 - t)\mathbf{X}_0 + t\mathbf{X}_1$



Training of velocity field (cont.)

Linear functional form of α_t and β_t

- α_t and β_t are set as

$$\alpha_t = t$$

$$\beta_t = 1 - t$$

- As a result, the objective function for $\mathbf{u}_t^\theta(\cdot)$ is simplified to be

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \mathbf{u}_t^{\text{target}}(\mathbf{X}_t) \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \frac{d}{dt} \mathbf{X}_t \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - (\mathbf{X}_1 - \mathbf{X}_0) \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta((1-t)\mathbf{X}_0 + t\mathbf{X}_1) - (\mathbf{X}_1 - \mathbf{X}_0) \right\|^2 \end{aligned}$$

Training procedure

Algorithm 3 Flow Matching Training Procedure

- 1: Input: A dataset of samples drawn from p_{data} , neural network u_t^θ
- 2: **while** not converge **do**
- 3: Set $\mathcal{L}(\theta) = 0$
- 4: **for** each mini-batch of data **do**
- 5: Select a sample x from the dataset
- 6: Sample a random time $t \sim \mathcal{U}[0, 1]$
- 7: Sample noise $\epsilon \sim \mathcal{N}(0, \mathbf{I}_d)$
- 8: Set $x_t = tx + (1 - t)\epsilon$
- 9: Compute loss
$$\mathcal{L}(\theta) = \mathcal{L}(\theta) + \|u_t^\theta(x_t) - (x - \epsilon)\|^2$$
- 10: **end for**
- 11: Update the model parameter via gradient descent

$$\theta \leftarrow \theta - \eta \frac{\mathcal{L}(\theta)}{d\theta}$$

- 12: **end while**
-

Facts about $\mathbf{u}_t^{\text{target}}(\mathbf{X}_t)$

- Let PDF of $\mathbf{X}_0$ be $p_{\text{init}} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{\text{init}}(\mathbf{x})$)
- Let PDF of $\mathbf{X}_1$ be $p_{\text{data}}(\mathbf{x})$ (i.e., $p_{\mathbf{X}_1}(\mathbf{x}_1) = p_{\text{data}}(\mathbf{x})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Let $\mathbf{u}_t^{\text{target}}(\mathbf{X}_t) = \frac{d\mathbf{X}_t}{dt}$ subject to $\mathbf{X}_0 \sim p_{\text{init}}(\cdot)$.

Suppose $\mathbf{u}_t^{\text{target}}(\mathbf{X}_t)$ is defined as above.

- It can be shown that $\mathbf{X}_t$, the solution to the ODE, transforms the distribution p_{init} at $t = 0$ to the data distribution p_{data} at $t = 1$.
- In other words, the probability path of $\mathbf{X}_t$ satisfy the boundary probability conditions.

See [strict proof](#) for Theorem 10 the manuscript

”An Introduction to Flow Matching and Diffusion Model”, by Peter Holderrieth and Ezra Erives.

Changes from FM to Guided FM

- Take the additional information y as input to the related quantities in an FM model.

FM	Guided FM
$\left(\begin{array}{c} \text{target} \\ \text{velocity} \end{array} \right): \mathbf{u}_t^{\text{target}}(\mathbf{x}_t)$	$\left(\begin{array}{c} \text{guided target} \\ \text{velocity} \end{array} \right): \mathbf{u}_t^{\text{target}}(\mathbf{x}_t y)$
$(\text{ model }): \mathbf{u}_t^{\theta}(\mathbf{x}_t)$	$\left(\begin{array}{c} \text{guided} \\ \text{model} \end{array} \right): \mathbf{u}_t^{\theta}(\mathbf{x}_t y)$
$\left(\begin{array}{c} \text{probability} \\ \text{path} \end{array} \right): p_{\mathbf{X}_t^{\theta}}(\mathbf{x}_t)$	$\left(\begin{array}{c} \text{guided} \\ \text{prob. path} \end{array} \right): p_{\mathbf{X}_t^{\theta} Y}(\mathbf{x}_t y)$

Training a guided FM

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $(\mathbf{X}_1, \mathbf{Y})$ be $p_{data}(\mathbf{x}, \mathbf{y})$ (i.e., $p_{\mathbf{X}_1, \mathbf{Y}}(\mathbf{x}_1, \mathbf{y}) = p_{data}(\mathbf{x}, \mathbf{y})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Objective function for $\mathbf{u}_t^\theta(\cdot)$:

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t, (\mathbf{X}_1, \mathbf{Y}), \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t | \mathbf{Y}) - \mathbf{u}_t^{\text{target}}(\mathbf{X}_t, \mathbf{Y}) \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, (\mathbf{X}_1, \mathbf{Y}), \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t | \mathbf{Y}) - \frac{d}{dt} \mathbf{X}_t \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, (\mathbf{X}_1, \mathbf{Y}), \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t | \mathbf{Y}) - \frac{d\beta_t}{dt} \mathbf{X}_0 - \frac{d\alpha_t}{dt} \mathbf{X}_1 \right\|^2 \end{aligned}$$

Training a guided FM

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $(\mathbf{X}_1, \mathbf{Y})$ be $p_{data}(\mathbf{x}, \mathbf{y})$ (i.e., $p_{\mathbf{X}_1, \mathbf{Y}}(\mathbf{x}_1, \mathbf{y}) = p_{data}(\mathbf{x}, \mathbf{y})$)
- **Train:** Objective function for training a guided FM model:

$$\arg \min_{\theta} \mathbb{E} \left(\begin{array}{l} t \sim \mathcal{U}[0, 1] \\ (\mathbf{x}_1, \mathbf{y}) \sim p_{data} \\ \mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{array} \right) \left\| \overbrace{\mathbf{u}_t^{\theta}(\mathbf{x}_t | \mathbf{y})}^{\text{DNN model}} - \mathbf{u}_t^{\text{target}}(\mathbf{x}_t | \mathbf{y}) \right\|^2 \quad \text{where } \mathbf{x}_t = \beta_t \mathbf{x}_0 + \alpha_t \mathbf{x}_1$$

$$= \arg \min_{\theta} \mathbb{E} \left(\begin{array}{l} t \sim \mathcal{U}[0, 1] \\ (\mathbf{x}_1, \mathbf{y}) \sim p_{data} \\ \mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{array} \right) \left\| \overbrace{\mathbf{u}_t^{\theta}(\mathbf{x}_t | \mathbf{y})}^{\text{DNN model}} - \frac{d\beta_t}{dt} \mathbf{x}_0 - \frac{d\alpha_t}{dt} \mathbf{x}_1 \right\|^2$$

- **Sample:** Generating new samples by solving the ODE:

$$d\mathbf{X}_t = \mathbf{u}_t^{\theta}(\mathbf{X}_t) dt, \quad \mathbf{X}_0 \sim p_{init}$$

Sampling from a guided FM

- **Sample:** Generating new samples by solving the ODE with an **input y** :

$$d\mathbf{X}_t = \mathbf{u}_t^\theta(\mathbf{X}_t|y)dt, \quad \mathbf{X}_0 \sim p_{init}$$

Remark: One can generate many data samples for the same input y ,

References

- Lecture notes by MIT
- Tutorial on Flow Matching for Generative Modeling

Questions?